

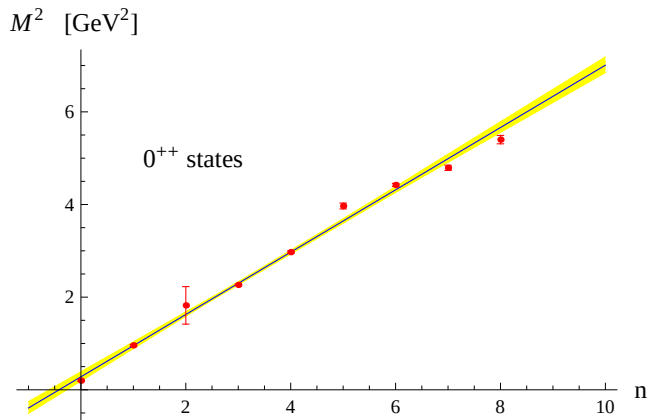


FIG. 1: (Color online) Radial Regge trajectory corresponding to the squared mass of all $J^{PC} = 0^{++}$ scalar-isoscalar states listed in the PDG tables [36]. The four heaviest 0^{++} states are not yet well established and are omitted from the PDG summary table. The error bars correspond to the errors in the determination of the square of mass, listed in [36]. The straight line is the result of Fit I with *half* the standard Regge slope, according to Eq. (2), with a and m_σ from Eq. (4). The band reflects the uncertainty in the parameters of Eq. (4).

fine-tuning condition on the σ mass is needed. Finally, in Section VII we come to our conclusions.

II. SCALAR-ISOSCALAR REGGE TRAJECTORIES

In this section the mass of the σ -meson is determined by extrapolating the upper part of the $J^{PC} = 0^{++}$ spectrum in the large- N_c Regge approach. In Ref. [37] the analysis of the radial and rotational Regge trajectories was carried out. In Ref. [38] the scalar sector was studied in a greater detail, including the modeling of the meson widths. Two parallel radial trajectories could then be identified, including three states per trajectory, which has resulted in rather poor statistics. In Fig. 1 we show the squared mass of all nine $J^{PC} = 0^{++}$ isoscalar states listed in the PDG tables [36], given for completeness in Table I. Note that the four heaviest 0^{++} states are omitted from the PDG summary table and in principle need confirmation. As we can see, an approximate straight line behavior looks quite appealing. To improve the statistics, we propose to fit with *half* the standard slope, thus including all the PDG scalar-isoscalar 0^{++} states together in the mass formula

$$M_S(n)^2 = \frac{a}{2}n + m_\sigma^2, \quad (2)$$

where $a = 2\pi\sigma$, and σ is the string tension. The same strategy was recently applied in Ref. [39].

In the first method, called Fit I, we minimize the χ^2

variable

$$\chi^2 = \sum_n \left(\frac{M_{f,n} - M_S(n)}{\Delta M_{f,n}} \right)^2. \quad (3)$$

We get, after *excluding* the lowest $n = 0$ state and using the 8 remaining PDG states, the central parameter values $a = 1.35 \text{ GeV}^2$ and $m_\sigma = 530 \text{ MeV}$, with $\chi^2/\text{DOF} = \chi^2/(8 - 2) = 9.9$.

A non-trivial issue, not addressed in the analyses of Refs. [37, 38], is the determination of errors of the parameters in the Regge fits. Actually, the large value of $\chi^2/\text{DOF} \sim 10 \gg 1$ found above prevents a reliable error determination of the model parameters a and m_σ and calls for corrections from different sources. A common way to overcome this difficulty is to rescale the weights in the χ^2 variable. In our case we simply replace $\Delta M_{f,n} \rightarrow 3\Delta M_{f,n}$, such that $\chi^2/\text{DOF} \sim 1$. In this case we get

$$a = 1.35(5) \text{ GeV}^2, \quad m_\sigma = 530(95) \text{ MeV} \quad (\text{Fit I}). \quad (4)$$

In the usual approaches (see, e.g., Refs. [38, 40]) a difference between the bare and physical pole masses, based on the K -matrix or the Breit-Wigner formula, is made. This generally introduces some model dependence and thus there may be systematic uncertainties in the masses related to the precise definition of the resonance parameters.

As we can see from Eq. (4), the fit is quite acceptable and implies the value of the string tension $\sqrt{\sigma} = 463(9) \text{ MeV}$. We note that the phenomenological value is $\sqrt{\sigma} = 420 \text{ MeV}$ from a global fit to the excited meson spectrum [37].

Formula (2) is equivalent to two parallel radial Regge trajectories with the *standard* slope

$$\begin{aligned} M_{S,-}(n)^2 &= a n + m_\sigma^2, \\ M_{S,+}(n)^2 &= a n + m_\sigma^2 + \frac{a}{2}. \end{aligned} \quad (5)$$

From here we get the mass-splitting formula

$$M_{S,+}(n)^2 - M_{S,-}(n)^2 = \frac{a}{2} = \pi\sigma \quad (6)$$

and, in particular,

$$2(m_{f_0(980)}^2 - m_\sigma^2) = a = 2\pi\sigma, \quad (7)$$

which works extremely well for the value $m_{f_0} = 980 \text{ MeV}$ and $m_\sigma = 530 \text{ MeV}$, yielding 1.36 GeV^2 for the l.h.s. vs. 1.35 GeV^2 for the r.h.s.

Given the fact that all states fit quite naturally into the pattern, we see no obvious way how glueball states could be singled out solely on the basis of belonging to one of the trajectories (5). Also note that a glueball corresponds to a bound state in gluodynamics, or equivalently QCD for infinitely heavy quarks (zero active flavors), while the states obtained here correspond to the limit of light quarks (two or three active flavors).